

Dr. Arnab Mapui



About Me

Innovation Skills
Critical Thinking
Subject Matter Expertise
Research Design

Personal

Arnab Mapui
Indian
[Google Scholar](#)
[Contact Me](#)

Research Interests

Nonlinear Dynamical Systems
· Lyapunov Stability · Impulsive
Control · Sliding Mode Control
· Event-Triggered Control

Interests

Sports (playing and following)

SHORT RÉSUMÉ

April 2026 – Present **Postdoctoral Fellow**
INDIAN INSTITUTE OF TECHNOLOGY KANPUR · Kanpur, India 📍

1em EDUCATION

2014 – 2017 **B.Sc**
MATHEMATICS · Burdwan University
Percentage: 69.38% 🏛️



2017 – 2019 **M.Sc**
MATHEMATICS · Visva-Bharati University
CGPA: 8.78/10 🏛️



2021 – 2025 **Ph.D.**
MATHEMATICS · IIT (BHU), Varanasi



TECHNICAL SKILLS

LyX

LaTeX

MATLAB

C/C++

Fortran

CERTIFICATES & ACHIEVEMENTS

2019 Qualified NET JRF (All India Rank: 93)
2020 Qualified GATE (All India Rank: 190)

LANGUAGES

Bengali	C2	Native
English	C2	● ● ● ●
Hindi	C2	● ● ● ●

PUBLICATIONS

1. **Practical prescribed/prescribed time stabilization via event-triggered impulsive control with actuation delays.** Systems & Control Letters. (Authors: **Arnab Mapui** & Santwana Mukhopadhyay)
2. **Lyapunov-like prescribed-time stability of impulsive systems via event & self-triggered impulsive control.** Nonlinear Analysis: Hybrid Systems. (Authors: **Arnab Mapui** & Santwana Mukhopadhyay)
3. **Lyapunov-like conditions for prescribed-time stability of perturbed impulsive systems.** Applied Mathematics and Computation. (Authors: **Arnab Mapui** & Santwana Mukhopadhyay)
4. **Prescribed-time stabilisation of nonlinear time-varying systems exhibiting impulsive behaviour.** International Journal of Control. (Authors: **Arnab Mapui** & Santwana Mukhopadhyay)
5. **On optimal upper bound for the settling time of fixed-time stable systems and its application in secure communication.** Nonlinear Dynamics. (Authors: **Arnab Mapui**, Md Arzoo Jamal & Santwana Mukhopadhyay)
6. **Fixed/predefined-time stability and designing of terminal sliding mode control of impulsive dynamical systems.** International Journal of Systems Science. (Authors: **Arnab Mapui**, Md Arzoo Jamal & Santwana Mukhopadhyay)
7. **Predefined-time stability and its applications using non-singular sliding mode control.** Communications in Nonlinear Science and Numerical Simulation. (Authors: **Arnab Mapui**, Md Arzoo Jamal & Santwana Mukhopadhyay)
8. **Further results on fixed-time synchronization of the memristor neural networks with impulsive effects.** Communications in Nonlinear Science and Numerical Simulation. (Authors: Md Arzoo Jamal, **Arnab Mapui**, Subir Das & Santwana Mukhopadhyay)
9. **On Optimal Fixed Time Stability: Its Application in Secure Communication Using Sliding Mode Control.** European Journal of Control. (Authors: Rajesh Kushwah, **Arnab Mapui** & Santwana Mukhopadhyay)

CONFERENCES & WORKSHOPS

- Presented a paper at the International Conference on Dynamical Systems, Control, and Applications, IIT Roorkee, July 2022.
- Attended NCMW Workshop, IISER West Bengal, Nov–Dec 2022.
- Participated in Annual Foundation School, NIT Durgapur, Dec 2022 – Jan 2023.
- Presented a paper at RAAM 2024, IIT (BHU), July 2024.
- Presented a paper at ICRTM 2025, University of Delhi, February 2025.
- Presented a paper at ICCAMSTIA 2025, Chennai, July 2025.

REFERENCES

- **Prof. Santwana Mukhopadhyay**
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- **Dr. Shyam Kamal**
Email: shyamkamal.eee@iitbhu.ac.in

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